

Short answer: **No** — you will **NOT** need infinite DB 😄

Long answer: you *do* need to manage data smartly, but what you're proposing (soft delete) is exactly how real systems scale.

Let's break your fear properly 📌



Your concern

"If I never delete labels, won't DB keep growing forever?"

- 👉 Yes, data grows
 - 👉 But **growth** ≠ **problem**
 - 👉 **Uncontrolled growth** = **problem**
-



Reality check (very important)

In your app:

How many labels can a user realistically create?

- 20–50 labels typical
 - Even heavy user: 200–300
- 👉 That's **tiny** for a database
-

Transactions?

- Let's say:
 - 5 transactions/day
 - 5 years usage
- 👉 $5 \times 365 \times 5 = \sim 9000$ transactions
- 👉 MongoDB handles **millions** easily
-



What actually causes DB explosion?

NOT soft deletes ❌

Real problems are:

- Storing images/videos badly
 - Logging too much
 - No cleanup strategy
 - Duplicate/unindexed data
- 👉 You are NOT doing these → you're safe
-

✓ Why soft delete is still correct

Because:

Approach	Result
Hard delete label	✗ Breaks analytics
Delete transactions	✗ Data loss
Remove label from tx	✗ Inconsistent insights
Soft delete	✓ Clean + safe

🔥 Real-world analogy

Think of:

- Bank statement

If you delete a category like:

👉 "Food"

Should bank delete all past food transactions?

✗ No

👉 They mark category as inactive

🧠 What real systems do (industry)

Apps like:

- Expense trackers
- CRMs
- E-commerce

👉 ALL use:

- Soft delete
- Archiving
- Background cleanup

✓ If you STILL worry about DB size

Here are **proper solutions** (not deleting blindly):

✓ 1. Archive old data

After 2–3 years:

JavaScript



Move to:

👉 transactions_archive collection

✓ 2. Periodic cleanup (safe delete)

Only delete:

- Labels not used in ANY transaction

```
const unusedLabels = await Label.aggregate([
  {
    $lookup: {
      from: "transactions",
      localField: "_id",
      foreignField: "labels",
      as: "tx"
    }
  },
  {
    $match: { tx: { $size: 0 } }
  }
]);
```

✓ 3. Limit user creation

Prevent spam:

JavaScript



```
    throw error  
  }
```

✅ 4. Indexing (you already did 👍)

❏ JavaScript



🚨 Important mindset shift

You're thinking like:

“How do I keep DB small?”

Start thinking like:

“How do I keep data CORRECT?”

👉 Because:

- Storage is cheap
- Wrong data is expensive

🏆 Final answer

- 👉 You do NOT need infinite DB
- 👉 Your data size is very manageable
- 👉 Soft delete is the **correct design choice**

💬 Honest advice

Right now your thinking is actually very good — you're considering:

- data integrity ✅
- performance ✅
- scalability ✅

👉 That's already beyond most students

If you want next step (this is where your project becomes 🔥):

👉 I can show you:

- How to build **analytics dashboard queries (group by labels, monthly trends)**
- OR optimize your schema for **fast reporting**

Just tell me 👍



later on we will do this
isDeleted and accordingly show stuff all that

for now i already have a lot on my hands

```
import Label from "../models/label.model.js";
import { createLabelSchema } from
"../validators/label.validation.js";
import mongoose from "mongoose";
import Transaction from "../models/transaction.model.js";
```

```
export const isValidObjectId = (id) => {
  return mongoose.Types.ObjectId.isValid(id);
};
```

```
// =====
// + CREATE LABEL
// =====
export const createLabel = async (req, res) => {
  try {
    const { value, error } =
createLabelSchema.validate(req.body);

    if (error) {
      return res.status(400).json({
        success: false,
        message: error.details[0].message
      });
    }
  }
```